Simultaneous measurement of the ν_μ charged-current inclusive cross section on hydrocarbon in five observables from MINERvA open data

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We report a simultaneous measurement of the ν_μ charged-current inclusive cross section in five observables, the muon transverse and longitudinal momenta, available hadronic energy, three-momentum transfer and hadronic invariant mass, unfolded without binning with OmniFold from MINERvA medium-energy open data and reported in sixteen cells with a covariance whose median per-cell uncertainty is 7.29%. Three external generator predictions compared on all sixteen cells give χ^2 of 109–272 for 16 degrees of freedom, and GiBUU, on 8 cells, 58. In a fake-data study, the reported uncertainty does not cover the estimator's bias at GiBUU-like truths or at a synthetic q_3 tilt. The same unfolding reproduces the integral of the published muon-kinematic cross section within about 1%, with a correlated shape difference.

I. INTRODUCTION

Multidifferential neutrino cross sections test interaction models across muon and hadronic observables. Their interpretation depends on correcting detector response and assessing the bias and uncertainty of the unfolding procedure. Iterative Bayesian unfolding on a binned migration matrix [1] is widely used, but the response tensor grows rapidly when additional observables are reported. OmniFold learns event weights in an unbinned feature space through classifier-based density-ratio estimation [2, 3]. This makes simultaneous higher-dimensional unfolding possible without specifying a separate binned response tensor for every projection. Unbinned unfolding has been applied to collider data, for example eight observables in deep-inelastic scattering at H1 [4] and twenty-four Z+jets observables at ATLAS [5]; see Ref. [6] for a survey. In neutrino physics it has been demonstrated on a public simulated T2K near-detector sample [7]. Multidifferential neutrino measurements on data reach three dimensions [8–11].

We apply the method to MINERvA medium-energy ν_μ data from the MINERvA open-data release [12]. The result is a simultaneous charged-current inclusive cross section in five observables, the transverse and longitudinal muon momenta p_T and $p_\parallel$, available hadronic energy E_{avail}, three-momentum transfer q_3, and hadronic invariant mass W, reported in sixteen cells with its covariance (Sec. V) and compared with five generator predictions (Sec. V B). We first reproduce the published double-differential cross section in transverse and longitudinal muon momentum, $d^2\sigma/(dp_T\,dp_\parallel)$ [13], as an end-to-end test of the unfolding, and we also give three- and four-observable central values. Reconstructed W is fixed by the reconstructed $(p_T, p_\parallel, q_3)$, so the five observables are simultaneous but not independent.

The validation studies assess the result through the two-dimensional reproduction, background-subtraction tests, recovery of generator-sized truth changes, a coverage test and a fake-data study of estimator bias. They examine whether the unfolding gives reliable multidimensional cross sections when the simulation mismodels the truth by generator-sized amounts. The coverage of unfolded intervals is the rate at which they contain the true cross section.

The result has three limitations, each stated with it. Interval coverage has not been validated. At two held-out GENIE v3 truths, a bias test finds the estimator's bias below the reported uncertainty in every cell, but does not test interval widths. In a fake-data study at 12 truths, the reported uncertainty does not cover the estimator's bias at the GiBUU-like truths or at a synthetic q_3 tilt (Sec. IV H). The generator χ^2 depends on the flux correlations: weakening only them by 10 % lowers the χ^2 of the sixteen-cell predictions by 24–48 % (Sec. V B).

Scope. The findings are those of one OmniFold implementation on one data set, for the observables, configurations, and deformations tested, rather than of multidimensional OmniFold in general. The three- and four-observable results are central values without intervals of validated coverage: the output of the joint-test estimator of Sec. III, with the observable definitions, simulated detector response, background treatment, and regularization given there. The sixteen-cell result of Sec. V also has total uncertainties, whose coverage is not validated. The companion note also gives simulation-calibrated tests of the generator predictions; they test hybrid null hypotheses, are calibrated within the implemented simulation model, and are not physical exclusions of the generators. They show no evidence of additional discrimination beyond matched coarse projections.

Section II describes the data, the simulation and the observables, and Sec. III the unfolding and the reporting

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cells. Section IV gives the uncertainties and the validation studies, from the reproduction of the published two-dimensional measurement to the fake-data tests of the sixteen-cell result, and Sec. V gives the result and its comparison with generator predictions. Section VI discusses what limits the estimator, and Sec. VI A summarizes the principal limitations. The companion analysis note [14] gives the full validation record, the construction of the uncertainties, and the tests with their assumptions, and is cited below by section.

II. DATA, SIMULATION, AND OBSERVABLES

The analysis uses the twelve MINERvA medium-energy, forward-horn-current run periods, corresponding to 1.057×10^{21} protons on target and an average neutrino energy near 6 GeV [13], from an earlier production of the MINERvA open-data release of pre-selected analysis ntuples (AnaTuples) [12] (see Data availability). The detector, including the downstream MINOS near detector that serves as its muon spectrometer, is described in Ref. [15]. The signal is a ν_μ charged-current interaction, with any final state, whose true vertex lies in the tracker fiducial volume ($5980 < z < 8422$ mm, inside a hexagon of apothem 850 mm) and whose muon has $\theta_\mu < 20°$, $1.5 < p_\parallel < 60$ GeV/c, and $p_T \leq 4.5$ GeV/c. Selected simulated events outside this definition, including charged-current ν_μ events whose muon lies outside these limits, are background. The selection requires a reconstructed vertex in the same volume and a reconstructed muon with $\theta_\mu < 20°$, matched to a MINOS track of negative charge, with a dead-time requirement, as in Ref. [13]. The flux treatment follows Ref. [16]. After selection, the sample contains 4.12 million selected data events. The simulation contains 32.8 million unweighted true signal events and 20.5 million selected simulated events. The simulation is scaled to the data exposure. The simulated events are generated with GENIE 2.12.6 [17] and the MINERvA Tune v1, which reweights GENIE with a non-resonant pion reduction, a random-phase-approximation correction, and an empirical low-recoil two-particle–two-hole (2p2h) enhancement [18]. Event selection is held fixed as the classifier inputs and unfolded observables grow from $(p_T, p_\parallel)$ to $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$. Adding features does not add selected events or relax the signal definition.

The unfolded observables are defined at truth level. Besides the muon momentum components $p_T = p_\mu \sin\theta_\mu$ and $p_\parallel = p_\mu \cos\theta_\mu$, they are the three-momentum transfer $q_3 = |\vec{q}|$ from the generator's four-vectors, the hadronic invariant mass $W = \sqrt{M^2 + 2M(E_\nu - E_\mu) - Q^2}$ with the struck-nucleon mass M, and the available energy

$$E_{\text{avail}} = \sum_p (E_p - m_p) + \sum_{\pi^\pm} (E_\pi - 135\,\text{MeV}) + \sum_{\pi^0} E_\pi + \sum_\gamma E_\gamma, \tag{1}$$

with a defined constant of 135 MeV, rather than the charged-pion mass, in the charged-pion term. All other final-state particles, including neutrons, contribute nothing. The sums run over the generator's final-state particles after final-state interactions, in which neutral pions appear undecayed. The 135 MeV constant is inherited from the analysis framework's definition. It defines the reported observable and is used consistently by the analysis and every prediction. Reconstructed E_{avail} is 1.17 times the calorimetric recoil energy in the tracker and electromagnetic calorimeter, with muon deposits removed. Reconstructed q_3 and W instead combine the muon kinematics with a separate calorimetric estimate q_0 of the energy transfer.

Reconstructed W is an exact function of the reconstructed $(p_T, p_\parallel, q_3)$, so the five reconstructed inputs share information and do not provide five independent constraints. The relation holds because $W^2_{\text{reco}} = (\bar{M} + q_0)^2 - q^2_{3,\text{reco}}$, with $\bar{M}$ an average nucleon mass, and $q_{3,\text{reco}}$ increases with q_0 at fixed muon kinematics. This relation is exact apart from a clamp at $W^2 = 0$. Only reconstructed W is fixed by the other inputs. Reconstructed E_{avail} comes from its own calorimetric estimator and is not a function of them. At truth level, θ_μ, and with it p_T and $p_\parallel$, is measured from a fixed beam axis rather than from the true neutrino direction. Truth W is therefore fixed by $(p_T, p_\parallel, q_3)$ only up to that event-by-event angle and the choice of M. The definitions are detailed in Ref. [14], Secs. 7.1 and 8.2.

The external predictions for CH are the untuned GENIE 2.12.10 without ("GENIE CV") and with ("GENIE + MEC") the Valencia 2p2h component [19], which models meson-exchange currents (MEC). The predictions for carbon are NuWro 21.09 [20] and GiBUU 2019 [21]. All are per nucleon of their target. The GENIE and NuWro predictions are reweighted exactly, event by event, to the analysis flux. The GiBUU sample stops at $E_\nu \approx 20$ GeV. The predictions compute W with the neutron mass for every event and measure p_T about their own neutrino axis. Hydrogen has no quasielastic, 2p2h, or coherent ν_μ charged-current channel, so normalizing per nucleon does not remove the difference between the targets. In GENIE's own CH sample, the carbon-only cross section per nucleon exceeds that of CH by 8.3 % for $W < 1.1$ GeV. The NuWro and GiBUU comparisons contain a target difference of this kind, which the comparison of Sec. V B does not correct. The comparisons of the predictions with the analysis definitions are given in Ref. [14], Secs. 8.1 and 8.3.

The fine grid is the analysis binning: the published 14×16 muon-momentum bins times 7 E_{avail}, 7 q_3, and 6 W bins. A coarse 3^5 grid in $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$ has its edges at the fine edges nearest the simulated tertiles. A coarse cell is kept if its expected reconstructed signal and its truth weight at the data exposure are both at least 1000 events. Of the coarse cells, 109 pass; these are the joint cells. The others, which hold 0.47 % of the expected signal, stay in the total only. The construction of the coarse grid is given in Ref. [14], Sec. 11.1.

III. CROSS-SECTION EXTRACTION

Each OmniFold iteration consists of two classification steps. A reconstruction-level classifier distinguishes data from simulated events and defines a likelihood-ratio weight. The event pairing transfers that weight to truth, including unmatched events, and a truth-level classifier propagates a smooth reweighting to the next iteration. Cross sections are formed by histogramming the reweighted truth events in the reporting binning and dividing by integrated flux, target count, and bin width. Events that fail reconstruction enter as unmatched truth events, so no separate efficiency correction is applied.

Backgrounds enter the reconstruction-level step through the density the classifier learns to match. In the purity-based subtraction every data event in reconstructed bin b carries the weight $w_{\mathrm{pur}}(b) = \max(0,\, 1 - B_b/D_b)$, where D_b and B_b are the data and predicted background counts in that bin, so the target density is $D(x)\, w_{\mathrm{pur}}(b)$. The unbinned negative-weight treatment instead adds simulated background events with negative weight to the data, so the target is $D(x) - B(x)$ pointwise; a classifier refinement [22], clipped at zero, converts the signed sample into non-negative weights before the first iteration. The two targets share bin integrals wherever $B_b < D_b$ but differ inside a bin wherever the background fraction varies within it (Ref. [14], App. C).

Each classifier is trained on the full sample whose weights it then sets, without a held-out or cross-fitted split, early stopping, or class balancing; the training details are in Ref. [14], Sec. 3.1.

Both estimators use five OmniFold iterations with gradient-boosted decision-tree classifiers [23, 24]. The iteration count and classifier capacity regularize the unfolding. The nominal settings are the scikit-learn defaults, with LightGBM matched to their capacity. The refinement capacity of the joint-test estimator was chosen in simulation closure studies. No hyperparameter search was performed on data.

Table I lists the two estimators and the results each is used for. The two-observable central value uses LightGBM at seed 42 with negative-weight background subtraction and a refinement of 400 trees and 31 leaves. This reproducible central value is that of its own systematic-universe ensemble, so its covariance is built with the same estimator. The three- and four-observable central values and the sixteen-cell five-observable result (Sec. V) use the joint-test estimator, a deterministic LightGBM with the same background treatment and refinement capacity, which removes most of the bias of purity-based subtraction at nominal truth in the highest-W cells (Sec. IV); the note's simulation-calibrated tests use the same estimator. The variants used in development, and the results that rest on them, are described in Ref. [14], Sec. 3.2.

The joint-test estimator keeps five iterations because 10, 15, or 20 iterations did not reduce its model dependence, measured as the largest residual proxy over the truth changes used in its development. The comparison and its values are given in Ref. [14], Sec. 9.4.

TABLE I. Estimators. Both use five OmniFold iterations, gradient-boosted trees of nominal capacity (100 trees, 8 leaves) in both OmniFold steps, and refined negative-weight background subtraction; the joint-test estimator is deterministic LightGBM (fixed seeds, reproducible training), and the two-observable estimator's refinement classifier is seeded at the estimator seed plus three. Results from the two estimators are not combined. Ref. [14], Sec. 3.2 gives the full matrix, including the variants used in development.

Estimator	Background treatment	Used for
joint-test estimator	negative weights; refinement 400 trees, 31 leaves	3D and 4D central values; sixteen-cell 5D result (Sec. V); held-out shapes (Fig. 3b); precision study
two-observable estimator: LightGBM, seed 42	negative weights; refinement 400 trees, 31 leaves	2D central value and covariance (Sec. IV B)

A. Reporting definitions and the precision target

To include model dependence, a dedicated study defines the envelope of a cell as the largest shift of its data result when the simulation's prior is reweighted to each of five generator-sized changes of the truth, and adds it linearly to the probabilistic half-width. The envelope is a sensitivity diagnostic that tracks the regularization bias, not an interval. Our precision target is a 10 % median half-width for the total 68% interval. With the envelope added, none of the reporting definitions of a five-observable measurement tested in the study meets this target. With the joint-test estimator, the envelope alone exceeds it for the 109 coarse joint cells and for the fallback definition, the 39 supported (E_{avail}, W) cells. For a coarser, non-nested partition of all five observables into 27 cells, the same estimator's envelope alone does not exceed it, but the 1.4 % normalization contained in every probabilistic half-width lifts the median above it. The study is described in Ref. [14], Sec. 9.4.

Over these definitions, the envelope has a median of about 10–13% per cell, comparable to the generator differences that such a measurement would need to resolve.

B. Reporting cells

A five-observable cross section with total uncertainties is reported separately from those reporting definitions: the joint-test estimator's data result on 16 cells, R00–R15. They come from the 32 boxes formed by one interior edge per axis (p_T 0.7, $p_\parallel$ 4.5 and q_3 2 GeV/c; E_{avail} 0.8 and W 1.8 GeV): each supported box with purity below 0.1 is merged into a neighbour, 4 boxes remain unreported, and every reported cell passes the support rule. The result's covariance is given in Sec. IV A.

IV. UNCERTAINTIES AND VALIDATION

A closure test measures how well a specified deformation of the known simulated truth is recovered, with the prior, the initial truth distribution, held fixed. Coverage, how often intervals contain their target over repeated experiments, tests the estimator and the interval construction together; the coverage tested here is pointwise, for each reported quantity separately, not the stronger simultaneous coverage of all of them. The closure, deformation, and coverage studies reweight the true signal of the simulated events used as pseudo-data, which changes their reconstructed and truth distributions together and keeps the detector response at fixed truth; the prior used to unfold them stays at Tune v1. The envelope of Sec. III A instead reweights the prior used to unfold the data. Every study uses the analysis simulation's detector response.

A. The reported covariance

The reported covariance of the sixteen-cell result is $C_{\text{total}} + C_{\text{resp}}$. Unlike the envelope of Sec. III A, its model term is a covariance built from three generator shifts. C_{total} sums the statistical, flux, interaction, detector, lateral and normalization uncertainties of the joint-test estimator and C_{model}. C_{model} is built from the change in the unfolded data when the prior is reweighted to GENIE + MEC, NuWro and GiBUU in turn; it represents unscaled generator-shape dependence, not a calibrated bound, and its coverage at truths held out of its construction has not been validated. C_{resp} is the response of the result to a one-standard-deviation scale of the reconstructed recoil energy in the simulation, for protons, mesons, electromagnetic and other deposits; it does not cover neutron or muon deposits, nor species, energy-dependence or resolution effects beyond a scale, and it enters only this result, its fake-data tests and its comparison in Sec. V B. The median per-cell relative uncertainty is 7.29% (6.74% without C_{resp}). The note gives the cells, the central values, the per-group uncertainties and the correlation matrix of the reported covariance.

B. Two-dimensional benchmark

The two-dimensional result is an end-to-end test of the unfolding against a measurement from the same data. Its integrated cross section is $3.069 \times 10^{-38}\,\text{cm}^2$/nucleon, compared with $3.039 \times 10^{-38}\,\text{cm}^2$/nucleon in the published analysis, a ratio of 1.010. The standardized bin differences formed with the published uncertainties have an RMS of 0.535. Against the full published covariance, the descriptive covariance-weighted distance is $\chi^2/\text{ndf} = 2.27$. It is computed with the pseudo-inverse over the 205 bins, which keeps all 205 eigenmodes (rank 205, condition number 1.5×10^{12}), so that ndf is that rank. It indicates a correlated shape difference, not a normalization difference, spread over many eigenmodes of the published covariance. The source of the shape difference has not been identified. Figure 1 shows the principal one-dimensional projections.

We quote the LightGBM result as the central value because its covariance uses the same estimator. The distance depends on the estimator (Ref. [14], Sec. 6.4). The learner was selected after the LightGBM and exact-split comparisons with the published result, so this benchmark is a reproduction rather than an independent validation. The purity-subtracted exact-split result differs from the central value by a median of 1.03 % per bin (maximum 17.6 %); see Ref. [14], Sec. 3.2.

C. Two-dimensional covariance

The covariance of our two-dimensional result is the sum of four blocks, each from LightGBM unfoldings with the central value's estimator: systematic universes (187 universes), statistical (300 bootstrap replicas), training seed (five seed settings, so that its own size is poorly determined), and run period (three period groups). The resulting relative uncertainty has a median of 6.41 % per bin and is 4.76 % on the integrated cross section, compared with 6.85 % and 4.61 % for the publication. The published flux uncertainty (median 4.01 %) is larger than ours (3.71 %), for a reason the released inputs do not let us determine. The covariance omits the approximation in the systematic-universe unfoldings for events that fail reconstruction, the learner dependence of Sec. IV B, and the dependence on the regularization; the dependence on the prior's generator shape is reported beside it (below). The covariance has not been independently validated. It provides no calibrated significance for generator comparisons. The block sizes, the assumptions of the block sum, and the tests of the approximation are given in Ref. [14], Secs. 6.3, 4.5, and 4.6.

The statistical block, with a median statistical uncertainty of 0.676 %, is a bootstrap of 300 replicas drawn at the central value's training seed, in which the background templates are also resampled. The coverage of this block and of the total covariance has not been tested. We make no coverage claim on independent simulated samples or simultaneously over all reported bins and projections.

Data from different run periods are not mutually consistent within the other uncertainties. One period group's unfolded $p_\parallel$ distribution, taken relative to its own simulation, differs from that of the earliest periods by a step of about 7 % near 8 GeV/c. The cause is not known, and these data cannot establish which period is correct. The run-period block, built from double ratios among the three period groups (each group's unfolded result over its own simulated truth, with no group chosen as the reference), includes this inconsistency in the combined result, 2.0 % on the integrated cross section. Because the block is built from the observed differences among the periods, its size is not validated and its own consistency check cannot fail.

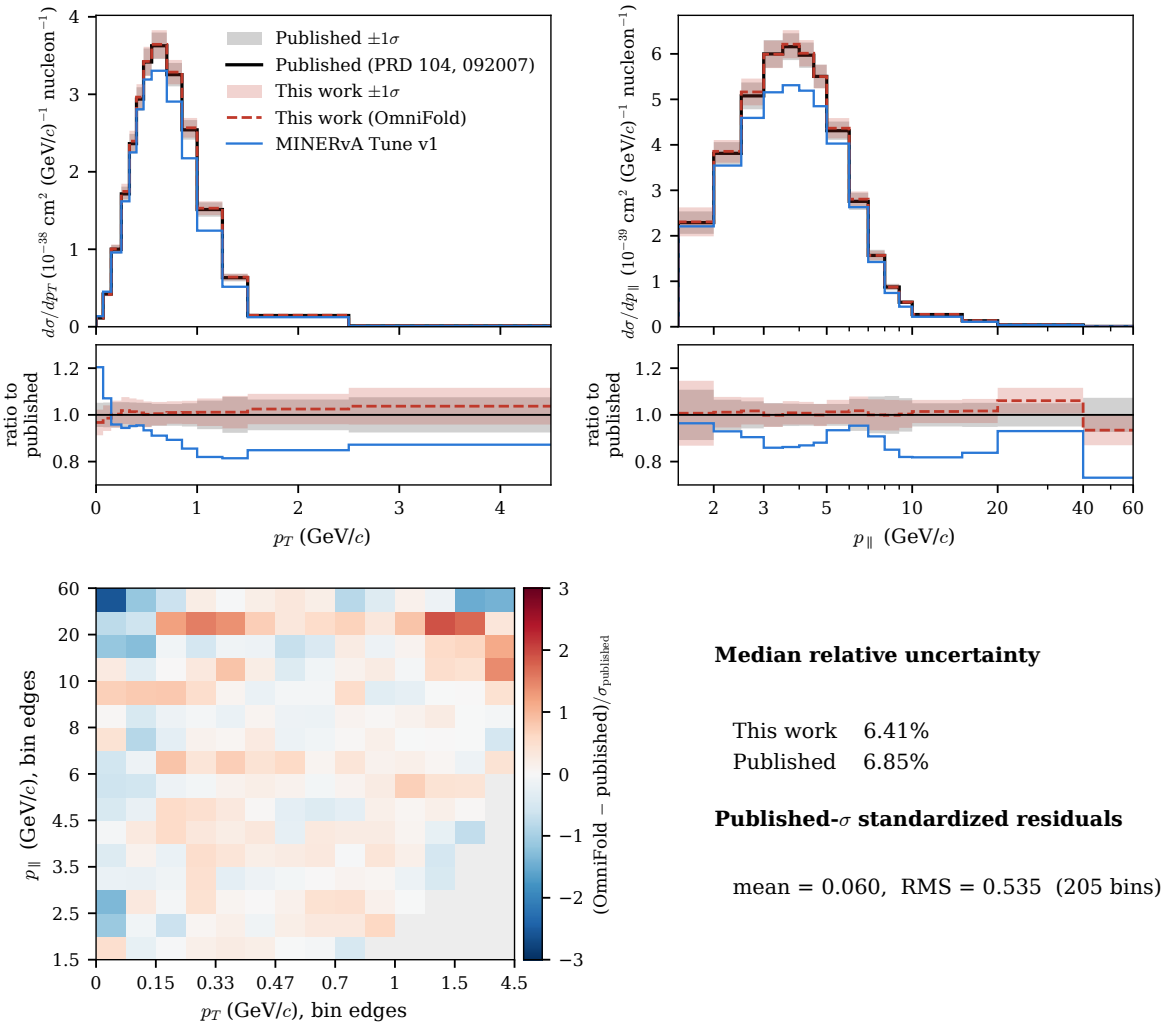


FIG. 1. Comparison with the published result [13]. Top: p_T and $p_\parallel$ projections of the published result and of this analysis, each with its total $\pm1\sigma$ band (this analysis: statistical, systematic-universe (including the normalization), training-seed and run-period blocks of the covariance, Sec. IV C; it omits the learner dependence and C_{model}, and the two bands are correlated, so their overlap is not a compatibility test), and the MINERvA Tune v1, with ratios to the published result below. Bottom left: (OmniFold − published)/$\sigma_{\text{published}}$ per reported bin, with cells drawn at equal size (grey: not reported). Bottom right: median relative uncertainties and residual summary.

We report the dependence on the prior's generator shape as a separate covariance beside the four-block covariance, $C_{\text{model}} = \sum_k s_k s_k^{\mathsf{T}}$. Here s_k is the per-bin relative change of the result when the prior's truth distribution is reweighted by the ratio of the $(p_T, p_\parallel)$ shape of generator k (GENIE with meson-exchange currents, NuWro, or GiBUU) to that of GENIE, with its normalization kept. Its median is 3.71 % per bin. It is a measure of generator-shape model dependence, not a calibrated bound on the unfolding bias. Added to the four-block covariance, it gives a total with a median of 7.90 % per bin. The comparison with the published uncertainty uses the covariance without C_{model}, because neither has a component for the residual dependence on the prior's generator shape: the published analysis chose its ten iterations in a fake-data study and assigns no uncertainty to the remaining dependence [13]. The iteration counts differ (ten there, five here), so we make no statement about the size of the published analysis's residual.

D. Normalization and closure

Of the joint-test estimator's results, only the three-observable marginal has been compared with the lower-dimensional result it extends: its area-weighted total agrees with the two-dimensional result to 0.141 %, a comparison across the two estimators. The normalization checks of the earlier purity-subtracted five-observable central value are described in Ref. [14], Sec. 8.1, and closure checks of its estimator in three and four observables in Ref. [14], Sec. 5.1.

Figure 2 shows that, with purity-based background subtraction in a deterministic LightGBM estimator, end-to-end pseudo-experiments at nominal truth reproduce the input except in the highest-W cells, which are biased by up to 4 % in simulation. In the same estimator, unbinned negative-weight subtraction with a refinement of 100 trees and 8 leaves reduces it to below 0.3 % in those cells (at most 0.4 % elsewhere in the projection), leaving up to 1 % in some joint cells. On data, the two treatments differ by at most 1.4 %. Both kinds of closure study are described in Ref. [14], Sec. 5.1.

In two observables, a closure test with backgrounds (20 pseudo-experiments at nominal truth) puts the purity and refined estimates more than one statistical standard deviation apart in 14 of the 205 cells. The purity estimate lies up to 5.9 statistical standard deviations from the truth (more than one total standard deviation in one cell), and the refined one lies within 0.44 in every cell. The two-dimensional result therefore uses the refined estimator (Ref. [14], App. C.6).

E. Generator-sized truth changes

Figure 3 shows that the unfolding recovers only part of a change of the E_{avail} shape as large as the difference between generators. When the true shape is reweighted by the GiBUU-to-GENIE ratio, the (E_{avail}, W) cells unfolded with a development variant of the joint-test estimator using a 100-tree, 8-leaf refinement deviate from it by several percent in the largest cells and by up to 74 % at high W, in the cell $0.8 \leq E_{\text{avail}} < 1.5\,\text{GeV}$, $W \geq 3\,\text{GeV}$. These deviations occur with or without background, so background subtraction alone does not explain the residual. We interpret the residual as regularization bias, because it persists over the tested iteration numbers and classifier capacities, although no single cause is established. For other fixed generator shapes tested with the joint-test estimator, it is 16 % to 31 % in the most affected (E_{avail}, W) cell and lies within the quoted total uncertainty in about half of those cells. Those uncertainties come from an earlier purity-subtracted five-observable covariance that fails its specified acceptance criteria (Ref. [14], Sec. 9.6). For the sixteen-cell result, the model term of its reported covariance is the change in the unfolded data when the prior is reweighted to each generator (GENIE + MEC, NuWro, GiBUU), unscaled and not a calibrated bound; Sec. IV H reports the fake-data tests of the estimator's bias. The deformation studies, the iteration and capacity tests, the exact binned comparison and the joint-test-estimator shapes are described in Ref. [14], Secs. 9.2 and 9.3.

In a binned, signal-only and noise-free diagnostic on the 109 joint cells and the truth changes used to develop the estimator, with the detector response fixed at the simulation's, an exact binned iterative unfolding with five iterations leaves about the same median residual as the boosted-tree estimator. It covers only the residuals whose diagnosis is stable under the truth resolution (38 % of the joint-cell residuals above 2 %). What unbinned information could recover, the coverage failure, and the validity of any interval lie outside its scope. Within that scope, the residual is consistent with the regularization of the inverse problem at this binning and prior, rather than with the tree approximation of the classifiers (Ref. [14], Sec. 9.3).

F. Coverage

Statistical intervals from a deterministic LightGBM estimator with purity-based subtraction, corrected for the estimator's closure bias at nominal truth, miscover at the tested truths, most strongly when the true signal rate is lowered by about 10 %. Under that change, 51 of the 153 reported cell and projection integrals have a 68% coverage whose upper confidence bound lies below the acceptance threshold of 0.66, with coverage as low as 0.06. The end-to-end test unfolds 400 pseudo-experiments per truth.

A diagnosis attributes the miscoverage to a residual bias that grows with W and survives the nominal-truth bias correction. The source of this residual bias has not been established. Agreement in the marginal normalizations coexists with substantial residual bias under the tested deformations. The interval construction and the diagnosis are described in Ref. [14], Sec. 9.5.

G. Estimator stability

The joint-test estimator is unstable at the level of rounding. Rounding-level changes of its inputs move its results by up to 2.6 statistical standard deviations, and those of its lower-capacity development variant by up to 2.5, so it fails a data-stability requirement of at most 0.3 (median) and 1.0 (maximum) statistical standard deviations. This spread lies inside the data-bootstrap statistical uncertainty, but it limits how exactly the central values can be reproduced.

H. Fake-data tests of the sixteen-cell result

To compare the estimator's bias with the reported covariance when the truth differs from the simulation, we unfold pseudo-data at 12 truths and compare the mean unfolded result with each truth on the 16 cells through χ^2 with 16 degrees of freedom. The truths are not held out: they are the simulation's prior; five hybrid truths, each a generator prediction on the fine grid with the MINERvA Tune v1 composition below it; NuWro shape changes in E_{avail} and in $(p_T, p_\parallel, E_{\text{avail}})$; a GENIE + MEC shape change in (E_{avail}, W); a GiBUU-to-GENIE E_{avail} ratio; a noise-free closure at the NuWro shape; and a synthetic tilt of the q_3 distribution at fixed (E_{avail}, W).

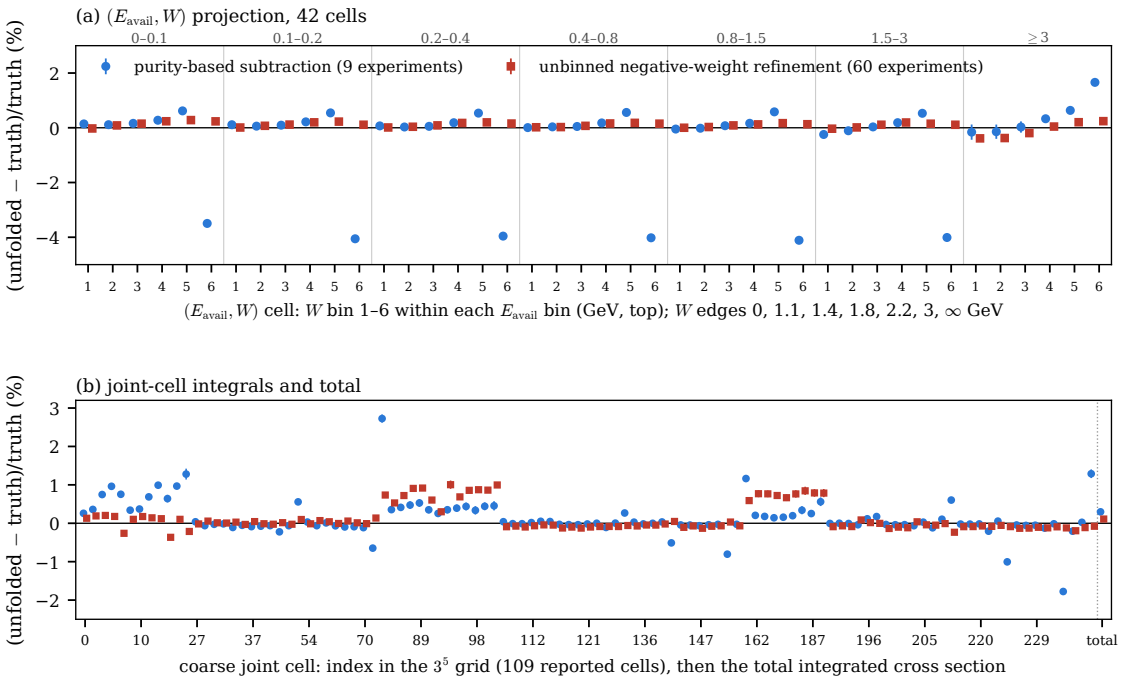


FIG. 2. Fractional closure residual, (unfolded − truth)/truth, at nominal truth, for pseudo-experiments that include backgrounds. Both from a deterministic LightGBM estimator. Circles: purity-based background subtraction (9 pseudo-experiments). Squares: unbinned negative-weight subtraction with a refinement of 100 trees and 8 leaves (60 pseudo-experiments). The pseudo-experiments are not paired. (a) The 42 (E_{avail}, W) projection cells, grouped by E_{avail} bin. (b) The 109 coarse joint cells, labelled by their index in the 3^5 grid, then the total integrated cross section. Error bars are the standard error of the mean.

For a truth from a generator that contributes to C_{model}, the covariance omits that generator's component, as the Cov. column of Table II shows. A truth is listed when $p < 0.01$ or when a cell has $|z| > 2$. Table II gives the results under the reported covariance and under C_{total} alone; 6 further truths are ungraded and given in the note.

Under the reported covariance, without the GiBUU component for the two GiBUU truths, the bias is listed at the GiBUU E_{avail} shape (cell R06, $z = 2.01$), at the GiBUU hybrid truth ($p = 8.0 \times 10^{-29}$; cells R02, R06, R07) and at the synthetic q_3 tilt ($p = 6.4 \times 10^{-6}$): the reported uncertainty does not cover the fake-data bias at these GiBUU-like truths or at the q_3 tilt. At the other 9 truths the bias passes both criteria: no cell beyond 2σ and $p \geq 0.01$.

Two GENIE v3 tunes, G18_10a_02_11b and AR23_20i_00_000, were generated for this test and held out of the analysis. Each reweights the simulated truth using its shape ratio to GENIE CV, constructed as for the C_{model} components. The simulated detector response is not probed. For each tune, 20 Poisson pseudo-experiments and a noise-free closure are compared with the truth using the full reported covariance. Both tunes satisfy the bias criteria: the pseudo-experiment means give $\chi^2 = 11.84$ ($p = 0.76$) and 16.86 ($p = 0.39$), respectively, with $|z| \leq 0.65$ in every cell of the means and closures. In cell R11 at G18_10a_02_11b, the bias lies below the reported uncertainty only with C_{model} included. The tunes share a quasielastic cross-section model; G18_10a_02_11b and the GENIE + MEC component of C_{model} also use Valencia-family MEC models. These tests provide partial evidence and are not independent. Omitting that C_{model} component gives $\chi^2 = 15.15$ ($p = 0.51$) and 24.60 ($p = 0.077$), respectively, as a descriptive comparison. The tests assess bias, not coverage: only event counts fluctuate, with systematic parameters fixed, so interval widths cannot be tested.

V. THE SIXTEEN-CELL CROSS SECTION

A. Result

Table III gives the sixteen-cell cross section: for each cell R00–R15, its box in the five observables, the cell-integrated cross section f, and its marginal relative uncertainty under the reported covariance $C_{\text{total}} + C_{\text{resp}}$ of Sec. IV A. The cells are correlated, so quantitative comparisons need the full covariance, whose correlation matrix is given in Ref. [14], Table 13; Table 11 there gives the uncertainties under C_{total} and C_{resp} separately.

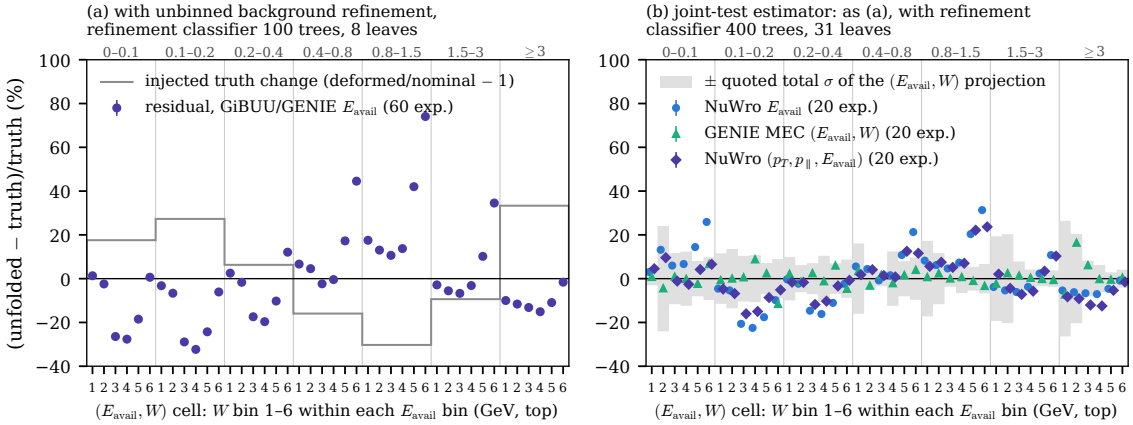


FIG. 3. Fractional recovery residual, (unfolded − truth)/truth, per (E_{avail}, W) cell, where the truth is each pseudo-experiment's deformed truth; all pseudo-experiments include backgrounds. (a) A development variant of the joint-test estimator, whose refinement has 100 trees and 8 leaves, with the truth reweighted by the GiBUU/GENIE E_{avail} shape ratio (60 pseudo-experiments); the grey step is the injected change of the cell truth. At nominal truth this variant's largest residual is 0.39 % (the squares of Fig. 2a). (b) The joint-test estimator, for three shape changes not used while it was chosen (NuWro in E_{avail}; GENIE + MEC in (E_{avail}, W); NuWro in $(p_T, p_\parallel, E_{\text{avail}})$), 20 pseudo-experiments each; the grey band is ± the total uncertainty of the (E_{avail}, W) projection of the covariance of an earlier purity-subtracted five-observable central value (Ref. [14], Sec. 9.6). At nominal truth (40 pseudo-experiments) its largest residual is 0.09 %. Neither seed set was held out from the estimator's development. W bin 6 is $W \geq 3$ GeV. Error bars are the standard error of the mean.

TABLE II. Fake-data tests on the sixteen cells: χ^2 (16 degrees of freedom), p and the cells with $|z| > 2$, against $C_{\text{total}} + C_{\text{resp}}$ and against C_{total}, each without the truth's own generator component where the second column says so (−).

		$C_{\text{total}} + C_{\text{resp}}$			C_{total}		
Truth	Cov.	χ^2	p	cells	χ^2	p	cells
simulation prior	full	0.00	1.00	—	0.00	1.00	—
Tune v1, hybrid	full	0.01	1.00	—	0.01	1.00	—
GENIE CV, hybrid	full	16.14	0.44	—	17.37	0.36	—
GENIE + MEC, hybrid	−GENIE + MEC	14.54	0.56	—	16.18	0.44	—
GENIE + MEC, (E_{avail}, W) shape	−GENIE + MEC	0.57	1.00	—	0.77	1.00	—
NuWro, hybrid	−NuWro	24.92	0.071	—	29.96	0.018	—
NuWro, E_{avail} shape	−NuWro	5.01	1.00	—	5.69	0.99	—
NuWro, $(p_T, p_\parallel, E_{\text{avail}})$ shape	−NuWro	30.83	0.014	—	36.19	0.0027	—
NuWro, closure	−NuWro	11.60	0.77	—	13.08	0.67	—
GiBUU, E_{avail} shape	−GiBUU	25.46	0.062	R06	35.75	0.0031	R06
GiBUU, hybrid	−GiBUU	175.13	8.0×10^{-29}	R02, R06, R07	246.75	$< 10^{-30}$	R02, R06, R07
synthetic q_3 tilt	full	53.42	6.4×10^{-6}	—	59.83	5.6×10^{-7}	—

B. Comparison on the sixteen reported cells

Unfolded results can be compared with any prediction at truth level without simulating each generator through the detector, which is what makes the two-dimensional result reusable, the higher-dimensional central values usable for qualitative comparison, and the sixteen-cell result usable for the comparison that follows. Table IV compares the sixteen-cell result of Sec. V with five predictions through χ^2 against the reported covariance plus each prediction's finite-sample (diagonal) covariance V_g, with shape-only χ^2 beside it. Figure 4 shows the data over each prediction, cell by cell. GiBUU's sample stops at $E_\nu = 20$ GeV, so it is compared on the 8 cells with $p_\parallel < 4.5$ GeV/c only. The reported uncertainty does not cover the fake-data bias at the GiBUU-like truths (Sec. IV H). NuWro and GiBUU are on carbon, with no correction to CH, and MINERvA Tune v1 is the simulation's prior. The GENIE + MEC, NuWro and GiBUU comparisons use covariance contributions derived from those generators; GENIE CV enters as their common reference.

Weakening only the flux correlations by 10 % lowers the χ^2 of the 16-cell predictions by 24–48 %. The same change to the detector or lateral correlations lowers it by at most 18 %. With the correlations of the interaction, flux, detector and lateral groups removed, every 16-cell prediction keeps $\chi^2 \geq 45$ for 16 degrees of freedom. The flux correlations are estimated from 100 hadron-production (PPFX) flux universes [16], weighted by the neutrino–electron and inverse-muon-decay constraint. Bootstrapping these universes, with the flux variances kept, gives 16th–84th

TABLE III. The sixteen-cell cross section: each cell's box (p_T, $p_\parallel$ and q_3 in GeV/c; E_{avail} and W in GeV), the cell-integrated cross section f (an integral over the box, not a differential density) in units of $10^{-40}\,\text{cm}^2$/nucleon, and its marginal relative uncertainty σ/f (%), the square root of the diagonal of $C_{\text{total}} + C_{\text{resp}}$ over f. The largest edge on each axis is the outer edge of the analysis grid, which contains every signal event (Ref. [14], Sec. 3.5). The last row gives the median over the cells. Quantitative comparisons need the full covariance (Sec. IV A).

cell	p_T	$p_\parallel$	E_{avail}	q_3	W	f	marginal σ/f
R00	0–0.7	1.5–4.5	0–0.8	0–2	0–100	32.6	7.10
R01	0–0.7	1.5–4.5	0–100	2–100	1.8–100	36.7	7.10
R02	0–0.7	1.5–4.5	0.8–100	0–2	0–1.8	6.31	14.00
R03	0–0.7	1.5–4.5	0.8–100	0–2	1.8–100	4.30	12.64
R04	0–0.7	4.5–60	0–0.8	0–2	0–100	49.0	4.20
R05	0–0.7	4.5–60	0–100	2–100	1.8–100	9.98	7.47
R06	0–0.7	4.5–60	0.8–100	0–2	0–1.8	5.57	5.89
R07	0–0.7	4.5–60	0.8–100	0–2	1.8–100	3.21	7.98
R08	0.7–4.5	1.5–4.5	0–0.8	0–100	0–1.8	13.9	6.25
R09	0.7–4.5	1.5–4.5	0–100	0–100	1.8–100	39.6	6.26
R10	0.7–4.5	1.5–4.5	0.8–100	0–2	0–1.8	7.83	8.59
R11	0.7–4.5	1.5–4.5	0.8–100	2–100	0–1.8	5.61	13.69
R12	0.7–4.5	4.5–60	0–0.8	0–2	0–100	22.0	6.63
R13	0.7–4.5	4.5–60	0–100	2–100	0–1.8	9.04	11.07
R14	0.7–4.5	4.5–60	0–100	2–100	1.8–100	49.6	5.27
R15	0.7–4.5	4.5–60	0.8–100	0–2	0–100	11.3	7.94
median							7.29

TABLE IV. χ^2 and its number of degrees of freedom of each prediction against the sixteen-cell result, under $C_{\text{total}} + C_{\text{resp}} + V_g$, with V_g the prediction's finite-sample (diagonal) covariance; shape-only χ^2 frees the normalization.

Prediction	χ^2 and ndf	shape-only
GENIE 2.12.10 CV	109/16	109/15
GENIE 2.12.10 + Valencia MEC	272/16	258/15
NuWro 21.09	188/16	188/15
GiBUU 2019	58/8	28/7
MINERvA Tune v1	330/16	311/15

percentile bands for each 16-cell χ^2, with edges within 6.2 % of the reported value and a median at most 1.0 % above it. These bands describe the scatter of the χ^2 from estimating the correlations with these universes, not its sensitivity to incorrect correlations, which the weakening probes; they are not intervals for the χ^2 under the true correlations.

C. Descriptive comparison of the central values

These comparisons use the three- and four-observable central values of the joint-test estimator, without uncertainties. GENIE, the MINERvA Tune v1, NuWro, and GiBUU all underpredict the unfolded central value at low E_{avail}, between the quasielastic and Δ peaks, the region of the established low-recoil discrepancy [18, 25]. Comparisons with pion final-state-interaction variations, GENIE's Valencia 2p2h component, and the published measurement are given in Ref. [14], Secs. 7.7 and 7.8.

The five-observable (E_{avail}, W) map of the earlier purity-subtracted central value, and its comparisons with these generators, are given in Ref. [14], Secs. 8.4 and 8.5.

VI. DISCUSSION AND CONCLUSIONS

The principal result is the sixteen-cell cross section of Sec. V, with its covariance and generator comparison.

Two validation findings point to the unfolded result retaining part of its prior. The first is the residual under a generator-sized change of the truth in Sec. IV E, which we interpret as regularization bias. The second is the envelope of Sec. III A, which tracks the same bias and alone exceeds the precision target in two of the three tested reporting definitions. The note's calibrated tests show the same pull: their null distributions for the external predictions are displaced upward by it, as the calibration reproduces (Ref. [14], Sec. 11.5). The miscoverage of Sec. IV F is also attributed to a residual bias, but whether that bias has the same origin is not established.

The studies of Sec. IV show what must be checked before unfolded intervals are interpreted: closure with the backgrounds included; truth changes of generator size, not only nominal closure; coverage under changes of the truth, not only at nominal truth; stability under training seeds and rounding-level input changes; and run-period consistency.

A. Limitations

The limitations of these results are stated with the results they qualify, in Secs. III–V, and those of the calibrated tests in the companion note. The principal ones are these. The two-dimensional covariance has not been independently validated, and its coverage has not been tested (Sec. IV C). The three- and four-observable central values lack intervals of validated coverage (Sec. V C). The

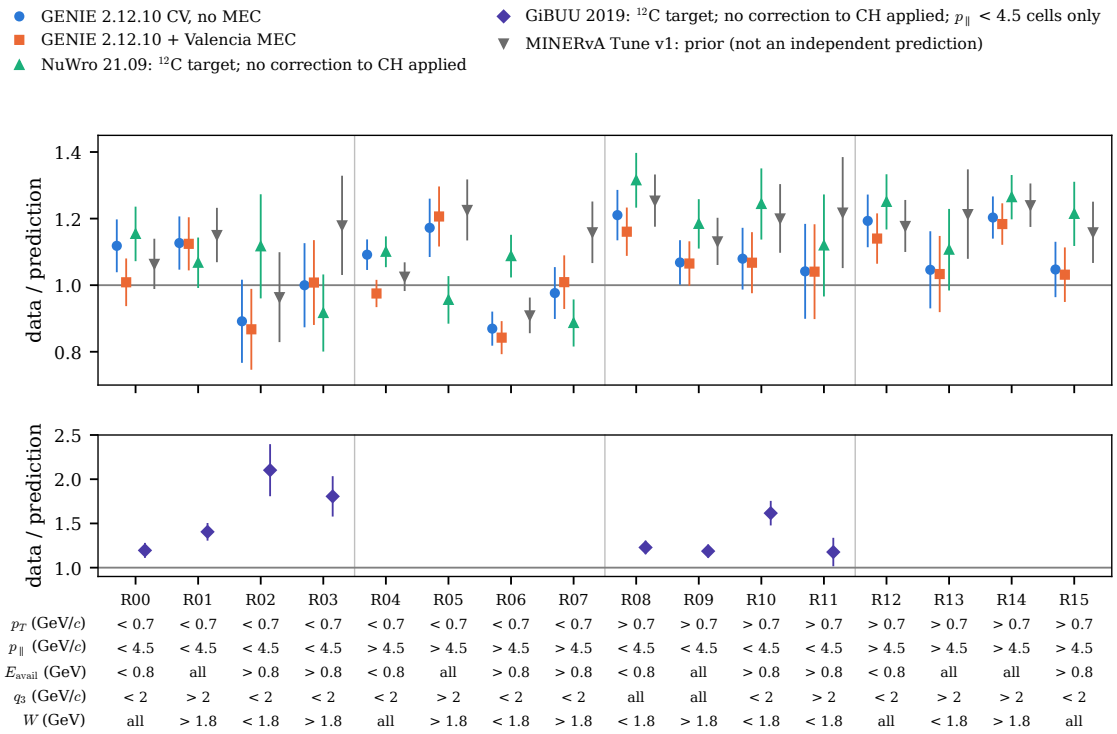


FIG. 4. Data over prediction, f/g, on the sixteen cells for the five predictions of Table IV. Each error bar is $(f/g)\,\sigma/f$, with σ the square root of the diagonal of $C_{\text{total}} + C_{\text{resp}}$; the predictions' finite-sample uncertainties are not drawn. GiBUU (lower panel) is shown on its 8 cells with $p_\parallel < 4.5$ GeV/c only. The key below the axes gives each cell's box. The χ^2 of Table IV uses the full covariance, whose correlations the error bars do not show.

sixteen-cell result also has total uncertainties, whose coverage is not validated; Sec. IV H reports the fake-data tests of the estimator's bias. The note's calibrated tests reject hybrid nulls, conditionally on the simulation's hadronic response, and the rejections carry over to the generator predictions themselves only under a bound on the residual below the fine grid, which has not been shown. Their calibration is valid to the extent that the simulated chain reproduces the analysis of the data (Ref. [14], Sec. 11.4).

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DATA AVAILABILITY

The input data and simulation are the MINERvA open-data AnaTuples, released under CC0 [12]. This analysis used an earlier production of that release; its 2,374 input files are identified by SHA256 checksum in the records cited in Ref. [14], App. G. A candidate release package of about 16 MB, not yet deposited, requires only numpy,

scipy, and matplotlib. It recomputes every p value, decision, robustness result, and power figure of the joint tests and the numbers quoted in Sec. V C, and it regenerates the analysis note's figures of the joint-test null distributions and the (E_{avail}, W) map. The other quoted numbers, except the multiplier scan, and the event-level processing require the archived analysis products, the open-data AnaTuples, and the generator samples produced for this analysis; Ref. [14], App. G states what each part of the release does and does not reproduce. [HOLD: deposit identifier and access route, after the separately authorized release decision.]

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